

Arkajyoti Mitra

[Email](#) | [LinkedIn](#) | [Google Scholar](#) | [Website](#)

Work Experience

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- Postdoctoral Researcher** April 2026 – Present
Deep Learning for Precision Health Lab (DLL / Montillo lab)
University of Texas Southwestern Medical Center (UTSW), Texas, USA
- Research focus: Develop foundation models for neuroimage analysis. Curating datasets for pre-training as well as designing optimization pipelines for downstream tasks such as segmentation and classification.
 - Collaborate and lead on multi-disciplinary projects, review and write research technical articles, and mentor graduates/undergraduates.
- Graduate Teaching Assistant / Research Assistant** Aug. 2020 – Dec. 2025
Cyber-Physical System Security (CSS) Lab
University of Texas at Arlington (UTA), Texas, USA
- Research focus: applying VLMs for autonomous driving and improving latent representations for seamless navigation.
 - Collaborate on projects, review and write research papers, and mentor undergraduate and master's students; strengthened communication, critical thinking, and time/resource management.
- Graduate Teaching Assistant** July 2018 – May 2019
Indian Institute of Technology (IIT) Dhanbad, India
- Assisted multiple instructors across varied courses and environments, gaining broad instructional experience.

Education

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- University of Texas at Arlington (UTA)**, Texas, USA Fall'20 – Fall'25
Doctor of Philosophy, Computer Science GPA: 3.9/4.0
- Indian Institute of Technology (IIT)**, Dhanbad, India Fall'17 – Spring'19
Master of Technology, Computer Science and Engineering CGPA: 8.65/10.0
- Kalyani Government Engineering College (KGEC)**, Kalyani, India Fall'12 – Spring'16
Bachelor of Technology, Computer Science and Engineering CGPA: 8.48/10.0

Research Interest

Medical Imaging, Foundational Model, Pre-/Post-training, Vision-Language Models (VLM), Gaussian Splatting, Flow-Matching, Diffusion, Generative Modeling, 3D Reconstruction, Multimodal Learning, Federated Learning, Reinforcement Learning, Autonomous Vehicular Perception Systems, Scene Understanding, Computer Vision Applications, Adversarial Attack, Systems Security, Network Security, Vehicular Security, Vision Security, LLM/VLM Security.

Publications

Conferences, Journals, and Workshops

- Paul Agbaje, **Arkajyoti Mitra**, Afia Anjum, P. Khose, E. Nwafor, Habeeb Olufowobi; **A Collaborative Distillation Framework for Graph Neural Networks**, *34th European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)*, pp. 673–678, 2026.
- **Arkajyoti Mitra**, Afia Anjum, Paul Agbaje, Mert D. Pesé, Habeeb Olufowobi; **FedVLM: Scalable Personalized Vision–Language Models through Federated Learning**, *28th European Conference on Artificial Intelligence (ECAI)*, vol. 413, pp. 4249–4256, IOS Press, 2025.
- **Arkajyoti Mitra**, Pedram MohajerAnsari, Afia Anjum, Paul Agbaje, Mert D. Pesé, Habeeb Olufowobi; **Beyond the Glow: Understanding Luminescent Marker Behavior Against Autonomous Vehicle Perception Systems**, *3rd USENIX Symposium on Vehicle Security and Privacy (VehicleSec'25)*, pp. 195–210, 2025.
- Afia Anjum, **Arkajyoti Mitra**, Paul Agbaje, Md Ahanaful Alam, Debashri Roy, Md Salik Parwez, Habeeb Olufowobi;

SemPerGe: Unveiling Text-based Adversarial Attacks on Semantic Communication, *IEEE Conference on Communications and Network Security (CNS)*, pp. 1–9, 2025.

- Aqsa Yousaf, **Arkajyoti Mitra**, Paul Agbaje, Afia Anjum, Habeeb Olufowobi; **DAPS-AGF: Depth-Aware Perceptual Similarity with Adaptive Gradient Filtering for Enhanced Outdoor Scene Reconstruction**, *IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pp. 2648–2656, 2025.
- Paul Agbaje, Afia Anjum, **Arkajyoti Mitra**, Habeeb Olufowobi; **Unveiling Graph Copycats: Inference Attacks with Student Models**, *Accepted in Privacy Enhancing Technologies Symposium (PETS)*.
- Aqsa Yousaf, Paul Agbaje, Afia Anjum, **Arkajyoti Mitra**, Habeeb Olufowobi; **Towards Enhanced Sparse-View Tomographic Reconstruction Using 3D Gaussian Splatting**, *Accepted in International Conference on 3D Vision*.
- Paul Agbaje, A. Mookhoek, Afia Anjum, **Arkajyoti Mitra**, Mert D. Pesé, Habeeb Olufowobi; **AutoWatch: Learning Driver Behavior with Graphs for Auto Theft Detection and Situational Awareness**, *2nd ISOC Symposium on Vehicles Security and Privacy (VehicleSec)*, 2024.
- Afia Anjum, Paul Agbaje, **Arkajyoti Mitra**, E. Oseghale, E. Nwafor, Habeeb Olufowobi; **Towards Named Data Networking Technology: Emerging Applications, Use Cases, and Challenges for Secure Data Communication**, *Future Generation Computer Systems*, vol. 151, pp. 12–31, 2024.
- Paul Agbaje, Afia Anjum, **Arkajyoti Mitra**, S. Hounsinnou, E. Nwafor, Habeeb Olufowobi; **Privacy-Preserving Intrusion Detection System for Internet of Vehicles using Split Learning**, *IEEE/ACM 10th International Conference on Big Data Computing, Applications and Technologies (BDCAT)*, 2023.
- Paul Agbaje, Afia Anjum, **Arkajyoti Mitra**, E. Oseghale, Gedare Bloom, Habeeb Olufowobi; **Survey of Interoperability Challenges in the Internet of Vehicles**, *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 12, pp. 22838–22861, 2022.
- Paul Agbaje, Afia Anjum, **Arkajyoti Mitra**, Gedare Bloom, Habeeb Olufowobi; **A Framework for Consistent and Repeatable Controller Area Network IDS Evaluation**, *Fourth International Workshop on Automotive and Autonomous Vehicle Security (AutoSec)*, 2022. **[Best Paper Award]**
- **Arkajyoti Mitra**, P. C. Tripathi, S. Bag; **Identification of Astrocytoma Grade using Intensity, Texture, and Shape Based Features**, *Soft Computing for Problem Solving: SocProS 2018*, vol. 1, pp. 455–465, 2019.

In Submission

- **Arkajyoti Mitra**, Kopil Sharma, Md Ahanaful Alam, Afia Anjum, Paul Agbaje, Md Salik Parwez, Ebelechukwu Nwafor, Habeeb Olufowobi; **RF-VLM: mmWave Aided Vision–Language Model for Autonomous Indoor Navigation**, *In Submission*.
- **Arkajyoti Mitra**, Paul Agbaje, Afia Anjum, Habeeb Olufowobi; **FedMoeFlow: Scaling Mixture-of-Experts Based Generative Models Through Federated Learning**, *In Submission*.
- **Arkajyoti Mitra** Md Sufyaan Saeed, Afia Anjum, Paul Agbaje, Habeeb Olufowobi; **SoK: Adversarial Analysis of Autonomous Vehicular Perception Systems**, *In Submission*.
- Paul Agbaje, Myriyam Ladhari, Jesse Chumo, **Arkajyoti Mitra**, Afia Anjum, Habeeb Olufowobi; **SoK: Security Strategies for Graph Neural Networks**, *In Preparation*.
- Afia Anjum, Paul Agbaje, **Arkajyoti Mitra**, Sena Hounsinnou, Md Salik Parwez, Habeeb Olufowobi; **Optimizing Resource Allocation for Multi-hop Sidelink Communication in 5G-NR**, *In Submission*.
- Afia Anjum, Aqsa Yousaf, **Arkajyoti Mitra**, Paul Agbaje, Megan Coffee, Habeeb Olufowobi; **Systematization and Empirical Evaluation of Explainable AI Methods for Clinical Decision Support Systems**, *In Submission*.
- Pedram MohajerAnsari, A. Domeke, J. de Voor, **Arkajyoti Mitra**, G. Johnson, A. Salarpour, et al.; **Discovering New Shadow Patterns for Black-Box Attacks on Lane Detection of Autonomous Vehicles**, *In Submission*.

Research Projects & Experience

Vision–Language Models (VLM)

Jan. 2024 – Dec. 2025

- Developing VLMs for multimodal reasoning and cross-sensor integration for unknown-terrain AV navigation.
- Developing a scalable framework for VLM training.
- Fine-tuning VLMs on VQA benchmark datasets (OK-VQA, TextVQA, GSM100K).

3D Reconstruction	May 2024 – Dec. 2025
• Building digital twin environments using 3DGS for safer development of autonomous driving. • Hyperparameter tuning for scene reconstruction; improving quality while reducing memory footprint.	
Generative Modelling	Jan. 2024 - Dec. 2025
• Building flow-matching models to learn velocity fields for efficient point-cloud generation. • Developing a scalable framework for diffusion-based models.	
Adversarial Analysis of Lane and Drivable Space Detection Models	Jan. 2024 – Aug. 2025
• Developed digital setup for simulation-based analysis of adversarial threats. • Built physical testbed for real-world analysis of adversarial threats. • Proposed a novel attack vector against SOTA models (YOLOv2, CLResNet, TwinLiteNet).	
MITRE eCTF Competition (Entered Design Phase)	Jan. 2025 – Apr. 2025
• Led a team to develop a secure satellite TV system using the MAX78000FTHR board.	
MITRE eCTF Competition (Entered Attack Phase)	Jan. 2024 – Apr. 2024
• Led a team to develop a secure medical device using the MAX78000FTHR board.	
CyberTractor Workshop	July 2022 – July 2022
• Secured second position by identifying/patching a security vulnerability on CAN via replay attacks.	
CyberTruck Workshop	June 2022 – June 2022
• Developed a spoofing attack on the CAN, compromising vehicular components (brakes, fuel indicators).	
Computer-Aided Identification of Astrocytoma Tumor Grades using Brain MRI	June 2018 – May 2019
• Developed a framework that detects and classifies brain tumors from MRI scans.	

Technical Skills

Programming & Software: Python, MATLAB, C++ , C, CARLA (simulator)

ML Frameworks: PyTorch, TensorFlow, Diffusers, Transformers

Awards and Scholarships

• Travel grant to attend PyTorch Conference'25	Aug. 2025
• Travel grant to attend VehicleSec'25	Aug. 2025
• Summer Research Fellowship at UTA	June 2025
• Travel grant to attend VehicleSec'24	Feb. 2024
• Travel grant to attend TAPIA'23	Sept. 2023
• Travel grant to attend CyberTractor Workshop	June 2022
• Travel grant to attend CyberTruck Workshop	July 2022
• Best Paper Award at AutoSec'22	Apr. 2022
• Second Best Paper Award, Computer Society of India (CSI), Kolkata	Aug. 2016

Other Activities

- Volunteered for VehicleSec'24
- Volunteered for SC'22